

Calculative Foundation Project Report

Vectors

- A vector is an ordered list of numbers representing a point or direction in space.
- Each student's subject scores form a vector with 5 elements.
- Vectors can be added, subtracted and scaled.
- Dot product of two vectors tells how similar they are in direction.
- Cross product gives a vector perpendicular to both input vectors.

Examples:

Example 1: Student 1 vector = [49, 77, 55, 60, 80]

Example 2: Dot product of [49,77,55] and [55,52,72] = $49 \times 55 + 77 \times 52 + 55 \times 72 = 11759$

Norms

- Norm measures the length or magnitude of a vector.
- L1 norm is the sum of absolute values of all elements.
- L2 norm is the square root of the sum of squared elements.
- Used in machine learning for regularization and optimization.
- Helps compare the overall magnitude of two student score vectors.

Examples:

Example 1: L2 norm of [3, 4] = $\sqrt{9+16} = 5$

Example 2: L1 norm of [49, 77, 55, 60, 80] = 321

Matrix Operations

- A matrix is a rectangular arrangement of numbers in rows and columns.
- Matrix addition combines two matrices element by element.
- Matrix multiplication follows the row-column dot product rule.
- Transpose flips a matrix along its diagonal swapping rows and columns.
- Determinant is a single value that tells if the matrix has an inverse or not.

Examples:

Example 1: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$

Example 2: Determinant of student score matrix = -3842650.0 (matrix is invertible)

Linear Transformations and Geometry

- A line is a 1D space and represents a single subject score like Math.
- A plane is 2D and is formed when we plot two subjects like Math and Science.
- A hyperplane is a higher dimensional space formed by all 5 subject scores.
- Adding more dimensions means adding more subjects to the representation.

- Dimensionality increases from 2D plane to 3D space to 5D hyperplane.

Examples:

Example 1: Math scores of 5 students form a line in 1D space

Example 2: Math and Science scores together form a 2D plane scatter plot

Eigenvalues and Eigenvectors

- Eigenvectors are directions in data that do not change when a matrix is applied.
- Eigenvalues are the scaling factors that show how much variance exists in each direction.
- Computed from the covariance matrix of the student score data.
- Higher eigenvalue means that direction explains more variance in the data.
- Used as the foundation for PCA and dimensionality reduction.

Examples:

Example 1: Solve $Ax = \lambda x$ for the covariance matrix of student scores

Example 2: Top eigenvector points in the direction of maximum score variance

LU Decomposition

- LU decomposition breaks a matrix into a Lower and Upper triangular matrix.
- A permutation matrix P is also produced to handle row swaps.
- Makes it easier and faster to solve systems of linear equations.
- More numerically stable than direct matrix inversion.
- Applied on the student score matrix to understand its internal structure.

Examples:

Example 1: Matrix $A = P \times L \times U$ where L is lower and U is upper triangular

Example 2: LU applied on 5x5 student score matrix gives L and U of same shape

Singular Value Decomposition (SVD)

- SVD decomposes any matrix into three matrices U , S and V^t .
- S contains singular values that show the importance of each component.
- Keeping only the top singular values reduces the data dimensions.
- Widely used in recommendation systems and image compression.
- Helps identify the most important patterns in the student dataset.

Examples:

Example 1: Student matrix decomposed into U (200x5), S (5,), V^t (5x5)

Example 2: Top singular value 892.3 captures the most variance in scores

PCA — Principal Component Analysis

- PCA reduces the number of features while keeping maximum variance.

- Finds new directions called principal components from eigenvectors.
- Projects the original 5-subject data onto 2 principal components.
- Makes it easier to visualize high-dimensional student data in 2D.
- Does not use class labels — it is an unsupervised technique.

Examples:

Example 1: 5D student data reduced to 2D using top 2 principal components

Example 2: First 2 components together explain around 64% of total variance

LDA — Linear Discriminant Analysis

- LDA finds the best axis to separate two or more classes in the data.
- Uses class labels to maximize the distance between class means.
- Classifies students into Above Average and Below Average categories.
- Unlike PCA, LDA is supervised and requires labeled data.
- Produces a single component that best separates the two student groups.

Examples:

Example 1: Students with average score above 70 are labeled Above Average

Example 2: LDA projects 5D score data onto 1D axis separating both groups

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